Table of Contents

HRAMDisplayDefTopic

<	>

About HRAM Display

About HRAM Display

The HRAM Display:

- Shows captured cycles, that is, pattern vectors as executed
- Provides a tabular view of captured HRAM data
- Includes fail overlay and actuals

To use the HRAM Display, run to a breakpoint in a test program. Use Tools>HRAMDisplay in the Launch group on the IG-XL Tools tab to bring up the display.

The HRAM Display uses HRAM for the data in its display.

The following topics are available:

- HRAM Setup
- Displaying Digital HRAM
- HRAM Display for Memory Patterns

IG-XL includes additional digital tools:

- PatternTool allows you to:
  - Create, edit, and debug pattern files

  - Overlay failures on programmed vectors
- Pattern Control Display:
  - Displays pattern information and memory consumption
  - Provides interactive pattern execution and management
- Waveform Display:
  - Graphically view waveforms over time and across pins of interest
  - Supports Protocol Aware (PA) frames and standard digital
  - Performs actual mode
- Protocol Definition Editor:
  - Define a protocol from scratch
  - Edit an existing protocol definition
  - Graphically view frame waveforms
- Protocol Studio:
  - Displays Protocol Aware (PA) frames at various levels of abstraction
  - Enables programming of PA port settings
  - Enables initiation of frames
- CyclePro:

- Raw cycle display mode
- Symbol10b display mode
- PCIe Link traffic decode (CMEM) and ATE Tx traffic decode (Tx HRAM)
- Layer Selection: Transaction, Data Link, PHY

For more information, see:

- About PatternTool
- Pattern Control Display
- About Waveform Display
- About Protocol Definition Editor
- About Protocol Studio
- About CyclePro

< >

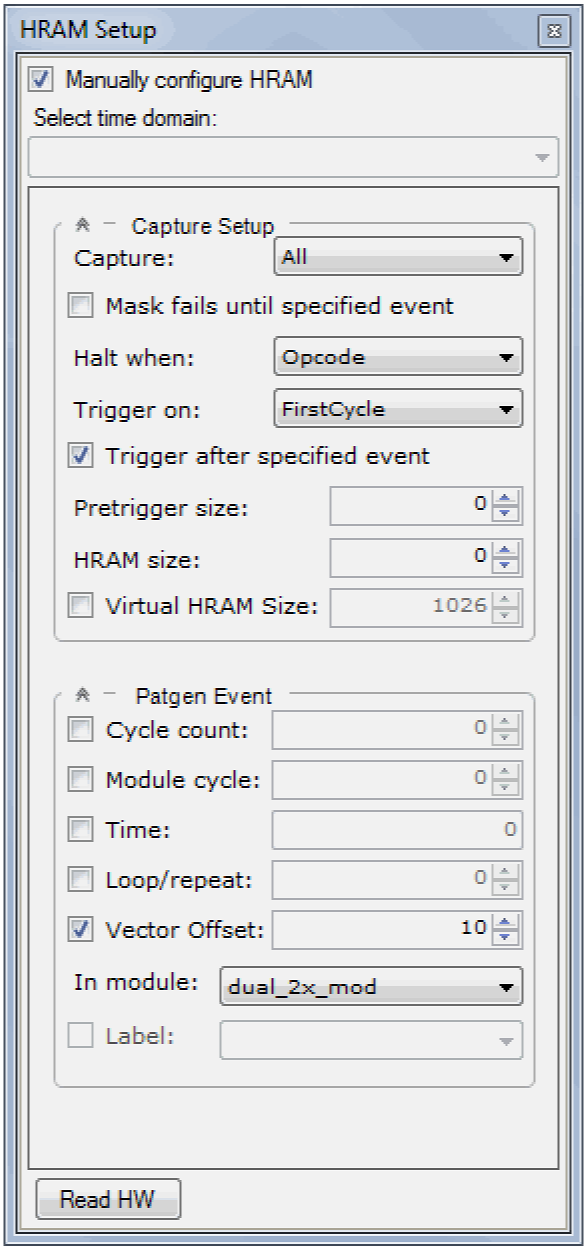
HRAMDisplayHramSetup

<	>

HRAM Setup

HRAM Setup

HRAM Display uses HRAM for the data in its display. This section describes the HRAM Setup pane in HRAM Display. To set up HRAM capture, use View>HRAM Setup to open the following dockable window:



HRAM Setup (Undocked)

To read the current hardware settings, click Read HW.

To change the settings, use the controls to specify the settings you want. Then click Apply.

Each event under Patgen Event and HPM Events has a check box. Use the check box to enable the event and the control to specify a numeric value. You must check the box before you can change the numeric value. You can disable an event by unchecking the box.

The HRAM settings are described in the following sections.

Capture Setup

Capture setup items specify what is to be captured.

Capture	Which cycles are to be captured once capturing is triggered. Select the capture type from the drop-down list:

	- All: All cycles - Fail: All failing cycles - STV: Cycles containing the control bit stv (store this vector) - FirstModFail: The first failing cycle in each pattern module - Pass: All passing cycles (Not supported on HPM and UltraPin4000) - FirstModPass: The first passing cycle in each pattern module (Not supported on HPM and UltraPin4000) - None: Do not capture See Notes on Capture Types. Note also that the only capture modes that are fully supported with keepalive when using multiple digital instruments are Fail and STV. You can still use the All capture mode while in keepalive, but the HRAM data may not accurately represent what happened during the pattern. As a result, do not use All with keepalive.
Mask fails until specified event	Whether to mask any failing vectors until the specified event occurs. Check this box if you want to mask fails until a specified event as defined in the Patgen Event group of parameters.
Halt when	Select when the PatGen will stop from the drop-down list: - Opcode: PatGen stops on the halt opcode in the pattern or on the implied halt at the end of a set of pattern modules in a single burst. Note that this mode is always enabled. If you select another halt mode, it is enabled in addition to the Opcode mode. - Fail: PatGen stops on first fail. Default option. Note that once the first fail is encountered, the PatGen does not halt for 255 major cycles, which is the pipeline depth for HSD1000, UltraPin800, UltraPin1600, UltraPin4000, and HPM.

	• HRAM Full: PatGen stops when HRAM has reached its size limit. The default is the size of the HRAM. However, you can program IG-XL to show a smaller HRAM size in the HRAM size parameter. • CMEM Full: PatGen stops when capture memory is full.
Trigger on	The cycle on which HRAM capture is to start. The options are: • First: Trigger on the first cycle of the pattern. • Fail: Trigger on the first cycle that fails. • Never: Do not trigger. See Notes on Trigger.
Trigger after specified event	Check this box if you want to trigger a capture after a specified event as defined in the Patgen Event group of parameters.
Pretrigger size	Set the number of HRAM cycles to view immediately before the trigger event. The pretrigger is programmable up to the size of HRAM. Note that the number of pretrigger cycles plus the number of posttrigger cycles (cycles captured after the trigger event) cannot exceed the size of HRAM.
HRAM size	Set up to the maximum number of cycles that the HRAM can capture. Leaving the value at 0 programs the maximum HRAM size. To see the maximum size for the current digital instrument, hover the cursor over this control. The screen tip displays the maximum size plus any requirement that the specified size must be a multiple of some value.
Virtual HRAM Size	Set up an HRAM capture that is larger than the maximum number of HRAM cycles that the instrument can physically capture. See Virtual HRAM Size. Virtual HRAM is supported only for the digital instruments (HPM, HSD1000, UltraPin800, UltraPin1600, and UltraPin4000).

Patgen Events

Patgen Event items specify the PatGen-related events that can trigger capture. This assumes that the Trigger after specified event box under Setup has been checked.

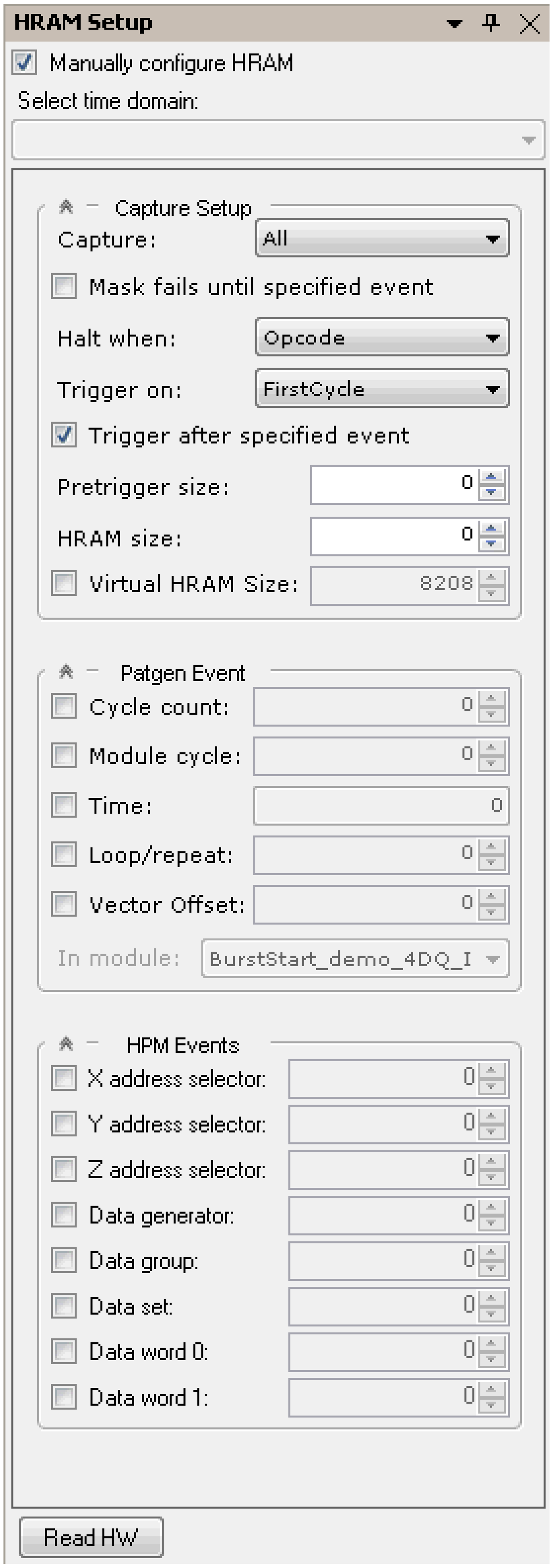
To enable a PatGen event, check its box and then enter the value for the event. You can enable any number of events. All enabled events must occur on the same vector to satisfy the PatGen event condition. This includes all enabled Patgen events and HPM events.

Cycle count	Set the number of cycles that must occur before the PatGen event. The PatGen will count all cycles and trigger capture once it reaches the specified number.
Module cycle	Set the number of cycles that must occur before the PatGen event. The PatGen will restart the count at the beginning of each pattern module and trigger capture when any module reaches the specified number.
Time	Set the number of time counts that must occur before the PatGen event. The time count is the same as the number of executed master oscillator (MOSC) cycles.
Loop/repeat	Set the number of pattern loop or repeat counts that must occur before the PatGen event. This option uses whatever counter is at the top of the stack for the cycle. For example, if a given cycle corresponds to a repeat inside of a loop, the repeat counter is on top of the stack and is therefore used for this option.
Vector Offset	Set the vector number of a specified pattern module that must be executed as the PatGen event. When specifying this PatGen event, you must use the In module field.
In module	Select the module the pattern module name that contains the vector offset number from the drop-down list.
Label	Select the pattern label that triggers the event from the drop-down list.

HPM Events

HPM Events items specify the HPM-specific events that can trigger capture. This assumes that the Trigger after specified event box under Setup has been checked. During pattern execution, values entered here are compared with the current hardware values, and a Sync is generated on each major cycle where the values match.

To enable an HPM event, check its box and enter a value for the event. You can enable any number of events. All enabled events must occur on the same vector to satisfy the PatGen event condition. This includes all enabled events in Patgen Event and HPM Events.



HRAM Setup with HPM Events

The meanings of the items are indicated by their names. The following table lists the valid range of values for each event:

Values for HPM Events

Option	Valid Values

X address selector	0 - 262143
Y address selector	0 - 65535
Z address selector	0 - 63
Data generator	0 - 15
Data group	0 - 3
Data set	0 - 3
Data word 0	0 - 4,294,967,295
Data word 1	0 - 4,294,967,295 Data word values are in decimal. They are usually specified in hex, with the range 0 - 0xFFFFFFFF.

Notes on Capture Types

Note also that the only capture modes that are fully supported with keepalive when using multiple digital instruments are Fail and STV. You can still use the All capture mode while in keepalive, but the HRAM data may not accurately represent what happened during the pattern. As a result, do not use All with keepalive.

When you set the capture type to Pass or FirstModPass (neither option is supported on HPM or UltraPin4000), IG-XL captures cycles on which any pin passes, even if other pins fail. These settings also cause the following conditions to respond to passing cycles in a similar way to how they normally respond to failing cycles:

- HRAM trigger on fail
- PatGen halt on fail
- CMEM capture fails
- Pattern fail flag

Note that when you set the capture type to Pass, the hardware only captures passing events when it performs a comparison between the programmed and actual values and that comparison passes. For instance, if you program the pin data state to X (don't care) for a vector, then you might expect the hardware to capture that vector since it will always pass, regardless of the actual DUT state data. In this case, however, if the actual DUT state data is 0 or 1, then the hardware will not capture that cycle because the X means IG-XL will not perform the comparison.

Notes on Trigger

Setting an HRAM trigger is how to start the countdown to halting storage into HRAM. That is, a trigger is a way of marking a certain event, such as a PatGen or fail event, as an event of interest that should be retained.

Note that the behavior for the FirstCycle and FirstFail depends on if you have checked the Trigger after specified event check box. If you are not waiting for a PatGen event (unchecked), then the behavior is as described above. If you are waiting for a PatGen event (checked), however, then these options have the following behavior:

FirstCycle	Trigger on the cycle when all of the specified events occur simultaneously.
FirstFail	Trigger on the first failing cycle on or after all of the specified events occur simultaneously.

When triggering HPM memory patterns, the HPM memory PatGen uses eight tracks of address and data information. Depending on the memory pattern, IG-XL selects address and data from a particular track. You can view which track is used for a given cycle in HRAM Display by enabling the Track check box. Since tracks 0-3 carry address and data and tracks 4-7 carry only address, IG-XL automatically enables and disables triggers on the eight tracks depending on the HPM events on which you choose to trigger.

If you do not check any of the data related events (data generator, data group, data set, data word 0, or data word 1), then IG-XL enables the trigger on all eight tracks. However, if you check HPM events on any of the data related events, or on a data event and an address event (X, Y, or Z address selector) simultaneously, then IG-XL disables the trigger on tracks 4-7 to prevent them from interfering with the data triggers on tracks 0-3.

<	>

HRAMDisplayMemoryPatterns

<	>

HRAM Display for Memory Patterns

HRAM Display for Memory Patterns

The HRAM Display shows captured cycles, that is, pattern vectors as executed, for algorithmic memory patterns.

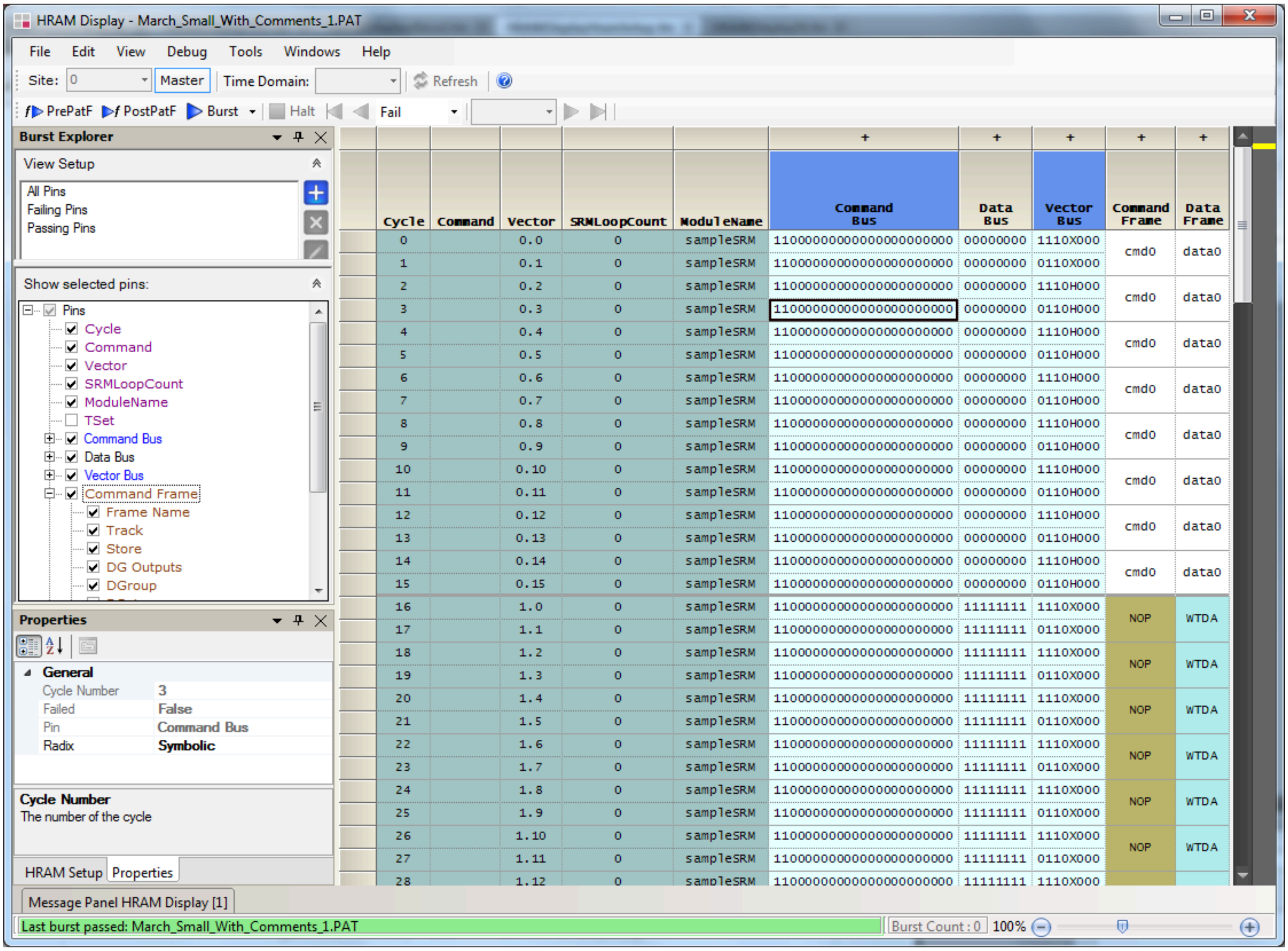
It provides a vertical digital display of pin data from captured cycles, plus MPG opcodes and PatGen opcodes.

Once a memory pattern has been loaded and run, you can debug the pattern using the familiar burst-setup trigger-reburst cycle to populate HRAM and to view the resulting HRAM capture to see what is right and wrong with the test program or the device.

To use the HRAM Display, run to a breakpoint in a test program. Select Tools>HRAMDisplay in the Launch group on the IG-XL Tools tab.

The HRAM Display uses HRAM for the data in its display.

HRAM Display shows a memory pattern like this:

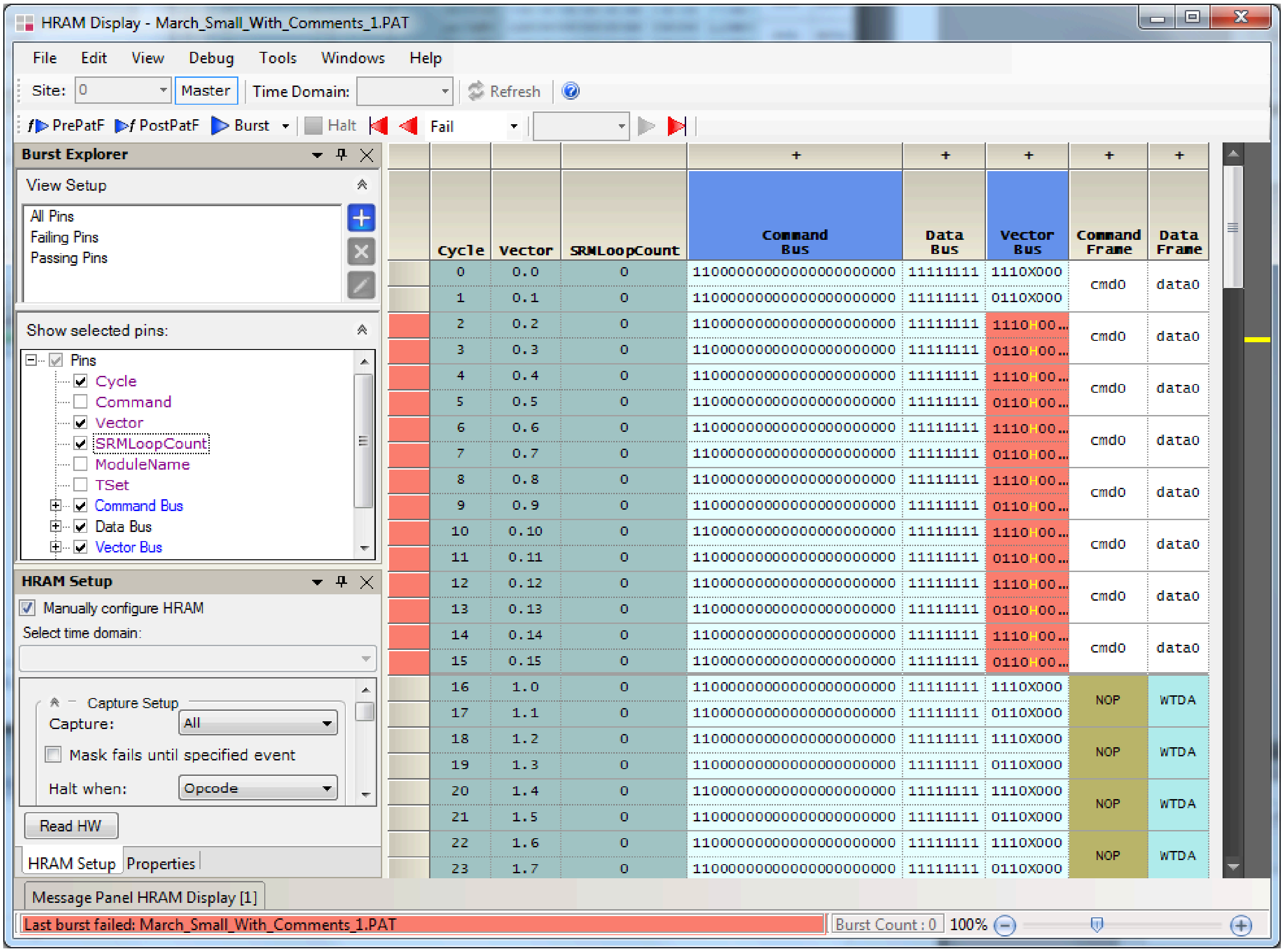


Memory Pattern in the HRAM Display

Each row represents a vector that has been applied by the PatGen hardware. Time increases in the downward direction.

Major vector boundaries are represented as thick dark gray lines that run across the entire display. Each minor cycle is displayed as a separate row of data in the grid.

Failing pin names and vectors have their topmost and leftmost columns colored pink. Cells at the intersection of a failing pin and vector are also in pink.



Memory Pattern with Fails

If a collapsed pin group or list contains a pin with fails, the entire name cell is colored pink. The cell with the fail is colored pink, with the failing data bit in yellow, as shown above in the first figure. In expanded mode, only the pin with the fail is colored pink.

Selecting Columns to Display

The Burst Explorer panel is a tree-like display that lets you choose which columns of data to show in the HRAM data grid. Checking the check box causes that column to be displayed.

Pin data column are listed under the node Pins. Failing buses and pins are indicated in red. Non-pin-related items such as PatGen opcode are displayed in purple. Also listed are nodes for Command Frame and Data Frame, which have check boxes for displaying the frames' parameters. You can also select or deselect the entire frame.

HRAM Display can display more data that is needed for a debug session. For example, when looking at Data frame data, the Data and Address Generator information is not very useful because the data and address information that relates to the Data frame appears earlier in the pattern. You could therefore hide these columns.

Cycle and Vectors

The Cycle column displays the number of this cycle. Each cycle increments by 1.

The Vector column shows the corresponding pattern vector number, indicating the major and minor vector: for example, 89.11 indicates the 11th minor vector of the 89th major vector.

With 16 minor cycles per major, 89.11 refers to the minor cycle 1435 (89*16+11). You can divide this minor cycle number by 8 to find the corresponding frame number in the UMC version of the pattern.

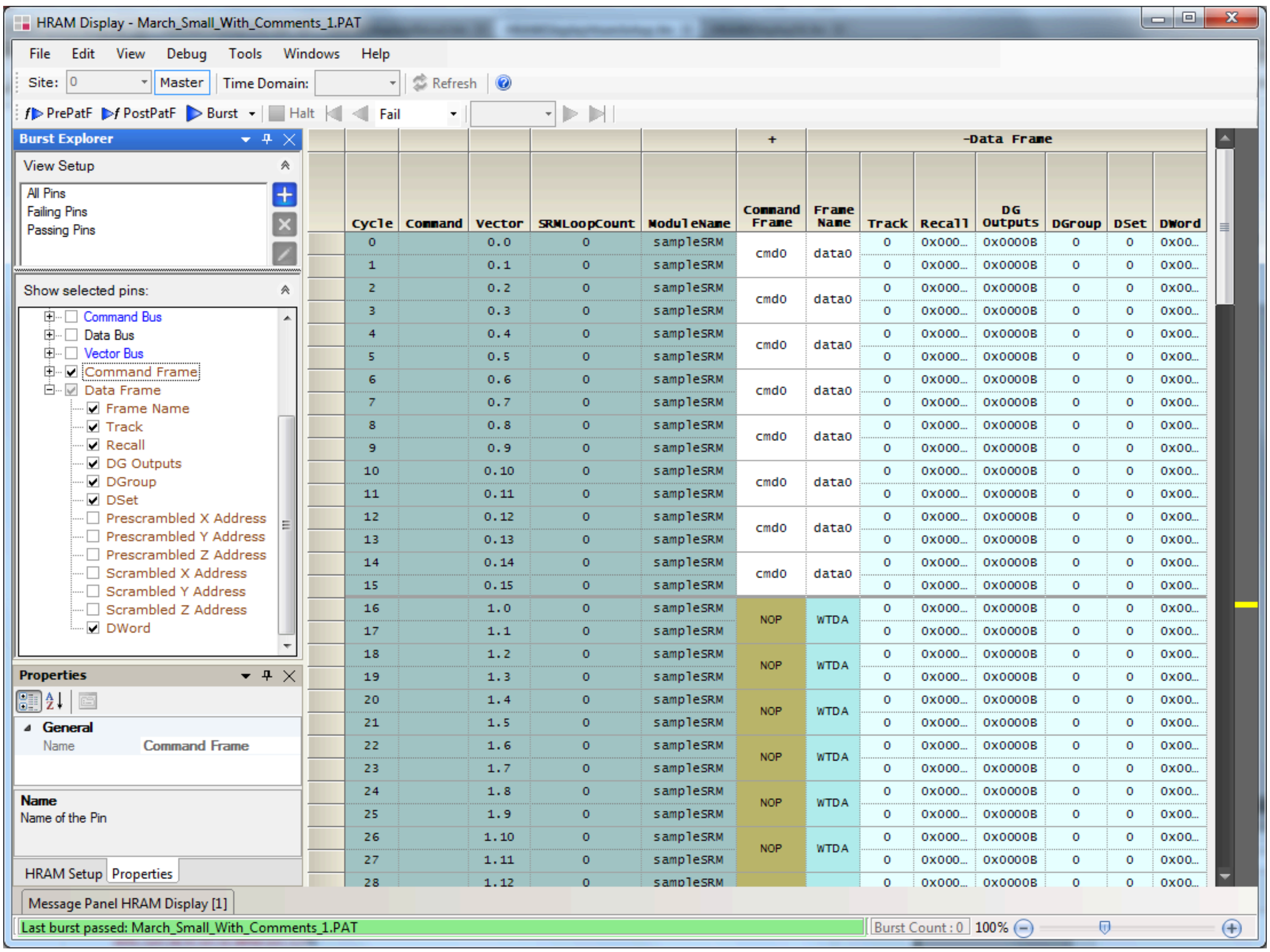
Pin Data

HRAM Display displays the pin data in the Command Bus, Data Bus, and Vector Bus columns. These columns can be expanded to show individual pins. The columns show the expected pin data. A red character indicates a fail. Failing pin names and cycles are indicated by the pink background in their cells.

To navigate in the display, you can use the First, Previous, Next, and Last buttons or commands.

MPG Data

HRAM Display displays MPG data in the Command Frame and Data Frame columns. You can expand these columns to show the data. The following shows the Data Frame columns expanded, but with only a subset of the individual columns selected for display:



HRAM Display, Columns Expanded

Each frame occupies two rows in the display, showing the data for UI 0 and UI 1.

Frames are color-coded based on the color assignments in the Frame Definition sheet. Color coding can be useful in showing where actions begin and end and how the Command and Data frames are coordinated. For example, in the display shown above, different colors are used in the Command Frame column for Nop and cmd0.

Note that these columns display what the MPG is doing for this cycle, not necessarily what is happening at the DUT. If latency is provided by Store/Recall (pattern latency), then the Data frame will be on a later row than the Command frame. In the display shown below, the RDA Command frame is seen near the top, but the corresponding RDDA Data frame appears eight frames (16 rows) later:

Cycle	Command	Vector	Loop Count	Module	Command Bus	DQ 0	DQ 1	DQ 2	DQ 3	DQ 4	DQ 5	DQ 6	DQ 7	Vector Bus	Command Frame	Data Frame
21536872		89.8	0	DDR3_...	0011000...	X	X	X	X	X	X	X	X	11110X...	ACT	DatNo...
21536873		89.9	0	DDR3_...	0011000...	X	X	X	X	X	X	X	X	00110X...		
21536874		89.10	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	11110X...	NOP	DatNo...
21536875		89.11	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	00110X...		
21536876		89.12	0	DDR3_...	0101000...	X	X	X	X	X	X	X	X	11110X...	RDA	DatNo...
21536877		89.13	0	DDR3_...	0101000...	X	X	X	X	X	X	X	X	00110X...		
21536878		89.14	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	11110X...	NOP	DatNo...
21536879		89.15	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	00110X...		
21536880		90.0	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	11110X...	NOP	DatNo...
21536881		90.1	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	00110X...		
21536882		90.2	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	11110X...	NOP	DatNo...
21536883		90.3	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	00110X...		
21536884		90.4	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	11110X...	NOP	DatNo...
21536885		90.5	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	00110X...		
21536886		90.6	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	11110X...	NOP	DatNo...
21536887		90.7	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	00110X...		
21536888		90.8	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	11110X...	NOP	DatNo...
21536889		90.9	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	00110X...		
21536890		90.10	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	11110X...	NOP	DatNo...
21536891		90.11	0	DDR3_...	0111000...	X	X	X	X	X	X	X	X	00110X...		
21536892		90.12	0	DDR3_...	0111000...	L	L	L	L	L	L	L	L	11110H...	NOP	RDDA
21536893		90.13	0	DDR3_...	0111000...	H	H	H	H	H	H	H	H	00110L...		
21536894		90.14	0	DDR3_...	0111000...	L	L	L	L	L	L	L	L	11110H...	NOP	RDDB
21536895		90.15	0	DDR3_...	0111000...	H	H	H	H	H	H	H	H	00110L...		

Pattern Latency

When latency is provided by delaying timing edges into future cycles (timing latency), the MPG address and data information shown in the Read command in the tool do reflect the Read address and data going to the DUT.

HRAMDisplayUsing.3.01

<	>

Displaying Digital HRAM

Displaying Digital HRAM

This section describes various aspects of displaying digital data in HRAM Display. The topics covered include:

- Viewing HRAM Data
- Navigation
- Burst Explorer and Properties
- Bursting Patterns

To bring up HRAM Display, on IG-XL Tools tab, in the Launch group, select Tools>HRAM Display. You can also bring up HRAM Display from other digital tools.

<	>

HRAMDisplayUsing.3.02

<	>

Displaying Digital HRAM : Viewing HRAM Data

Viewing HRAM Data

This section describes how to view HRAM data:

<	>

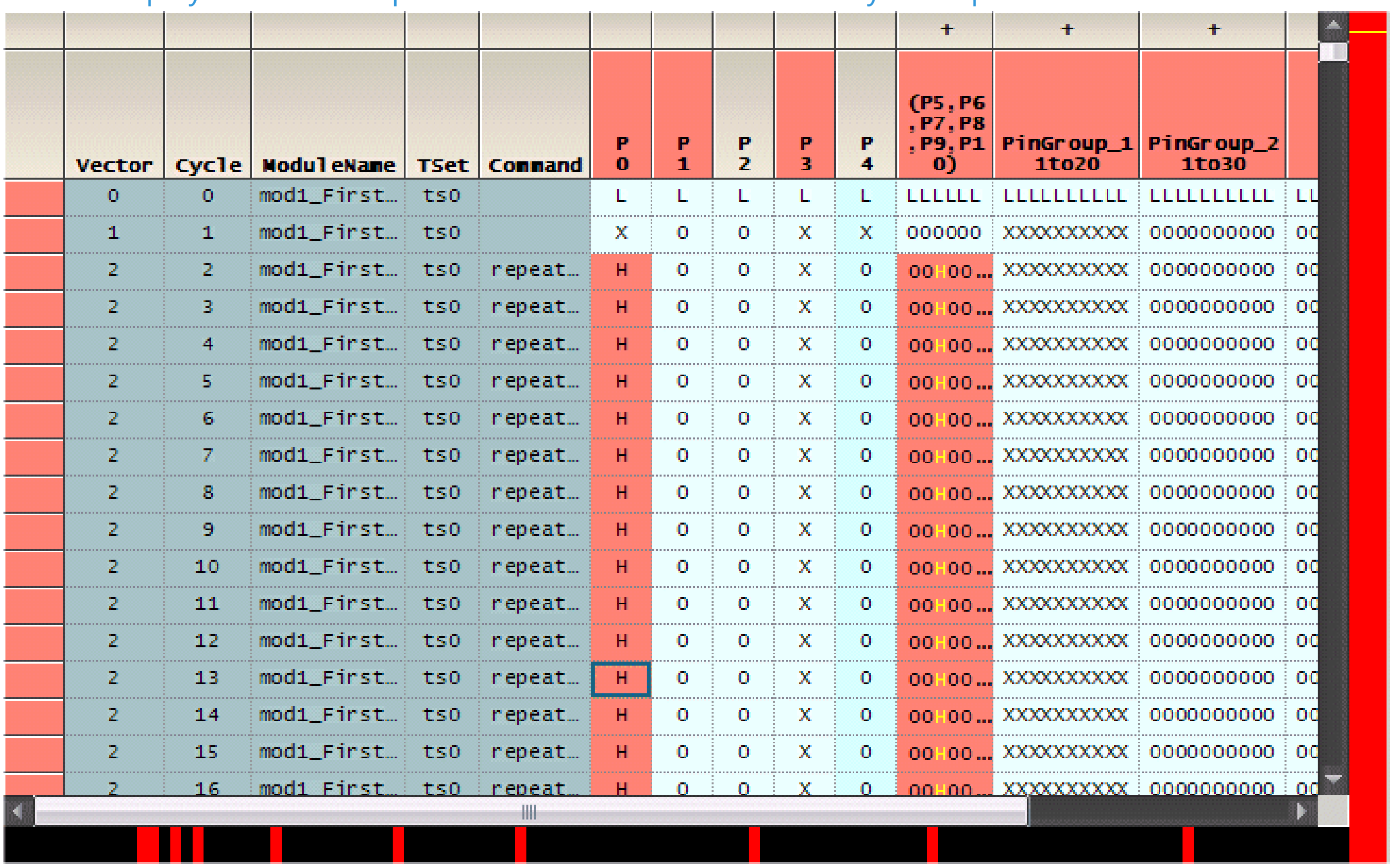
HRAMDisplayUsing.3.03

< >

Displaying Digital HRAM : Viewing HRAM Data : Display Basics

Display Basics

HRAM Display shows the captured data from the most recently burst pattern:



HRAM Display Showing Captured Data

The columns on the right side contain pin data. Each row represents a vector that will be applied by the PatGen hardware. Time increases in the downward direction. The vector number is a read-only field. The tool automatically increments the value from 0 in each module. In the case of scan vectors, the format is vectornumber.scanindex.

You can specify which PatGen data and pin columns are to be displayed using the Burst Explorer. See Burst Explorer and Properties.

Horizontal and vertical scrollbars are presented as needed if there is more information than can be displayed within the viewing area.

Failing pin names and vectors have their topmost and leftmost columns colored pink. Cells at the intersection of a failing pin and vector are also in pink.

Major vector boundaries are represented as thick dark gray lines that run across the entire display.

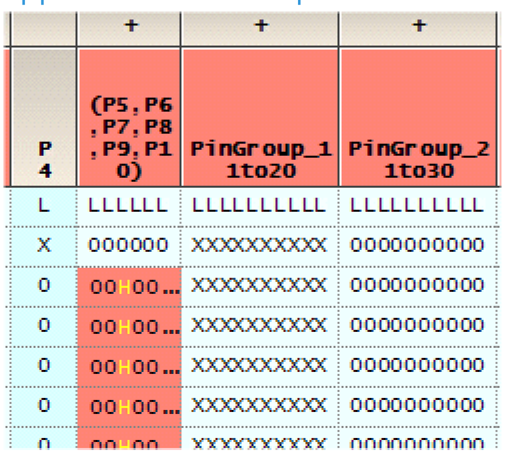
Module boundaries are represented as thick horizontal blue lines that run across the entire display, between the last vector of a module and the first vector of the next module.

Programmed and Actual Expect Data

You can specify whether HRAM Display shows programmed data or actual data. Use View>Expect Data to select programmed or actual data.

Displaying Pins, Groups, and Lists

To display as much data as possible, pin names are displayed vertically at the top of the pin data column. The names of pin groups and lists (a pin list name is a parenthesized list of the pins) appear in the same space:



Pin Names in the Display

You can expand a pin group or list by clicking the plus sign (+) above its name or compress it by clicking the minus sign (-):

	-(P5,P6,P7,P8,P9,P10)						-PinGroup_11to20							
P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	P15	P16	P17	P18
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L

Pin Group Expanded

If a collapsed pin group or list contains a pin with fails, the entire name cell is colored pink. The cell with the fail is colored pink, with the failing data bit in yellow, as shown above in the first figure. In expanded mode, only the pin with the fail is colored pink.

Label Data

By default, the Label column is not displayed. You can have it displayed by checking on the Label item in the Burst Explorer. See Selecting Pins and PatGen Data.

Note that the Label column displays only global labels. It does not display local labels.

Scan Data

HRAM Display shows scan data like this:

	Vector	Cycle	ModuleName	TSet	Command	P0	P1	P2	P3	P4	P5
	9	9	mod1_First…	ts0		L	L	L	L	L	L
	10	10	mod1_First…	ts0		L	L	L	L	L	L
	11	11	mod1_First…	ts0		L	L	L	L	L	L
	12	12	mod1_First…	ts0		L	L	L	L	L	L
	13	13	mod1_First…	ts0		L	L	L	L	L	L
	14	14	mod1_First…	ts0		L	L	L	L	L	L
	14:1	15	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:2	16	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:3	17	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:4	18	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:5	19	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:6	20	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:7	21	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:8	22	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:9	23	mod1_First…	ts0	repeat…	L	L	L	L	L	L
	14:10	24	mod1_First…	ts0	repeat…	L	L	L	L	L	L

Scan Data in the Display

A scan chain is indicated in the Vector column as m.n, where m is the vector with the scan opcode and n is the number of the vector in the scan chain.

Resetting Column Width and Hiding Columns

Commands in the Window menu let you modify column displays. Select a column and then select one of these commands from the Windows menu or from the right-click context menu:

Reset Column Width	Resizes the width of the column so that it displays the widest data cell of the column. By default, the column width is determined by the column header.
Hide Column	Hides the currently selected column. This unchecks the column in the Burst Explorer. To unhide the column, recheck the box in the Burst Explorer.

Using Zoom

There is a zoom track bar at the bottom of the window:

Zoom Track Bar

The horizontal track bar lets you increase or decrease the horizontal time scale of the display. Only the horizontal time scale is affected, not the vertical height of the lanes. The vertical track bar lets you increase or decrease the vertical voltage scale.

You can grab and drag the slider or click the track bar to move the slider in that direction.

<	>

HRAMDisplayUsing.3.04

<	>

Displaying Digital HRAM : Viewing HRAM Data : Taking and Viewing Snapshots

Taking and Viewing Snapshots

HRAM Display lets you take snapshots of the current state of the tool. Later you can view the snapshots for a visual comparison of the tool states.

The command View>Take Snapshot takes a snapshot of the HRAM Display window. The View>View Snapshot command lets you view the snapshots. It has these options:

Most Recent	Displays the most recent snapshot.
Other	Brings up a browser to let you select the snapshot to view.

The display uses the default Windows image viewer.

By default, the snapshot is saved to the test program folder. If the default folder is read-only, it is saved to the tmp folder. You are not prompted to specify a folder. The snapshot is saved as a JPEG file with a filename of this form: HRAM Display__YYMMDDHHMMSS.jpg. This means that the snapshots in the browser will be in chronological order.

<	>

HRAMDisplayUsing.3.05

<	>

Displaying Digital HRAM : Viewing HRAM Data : Selecting the Site and Time Domain

Selecting the Site and Time Domain

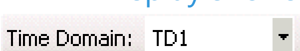
HRAM Display shows the data for one site at a time. Use the site selector to specify the site:

Site Selector

If you click Master, HRAM Display uses the master site as it is defined in DataTool. The site number displays the current master site but is grayed out because you cannot change it. If the master site changes, HRAM Display indicates that you need to refresh the display.

If you disable Master, you can manually select the site to be displayed. If you change the site, HRAM Display autorefreshes to display the data from the new site.

HRAM Display shows the data for one time domain at a time. Use this control:



Time Domain Control

If there is a single time domain, the control is disabled.

<	>

HRAMDisplayUsing.3.06

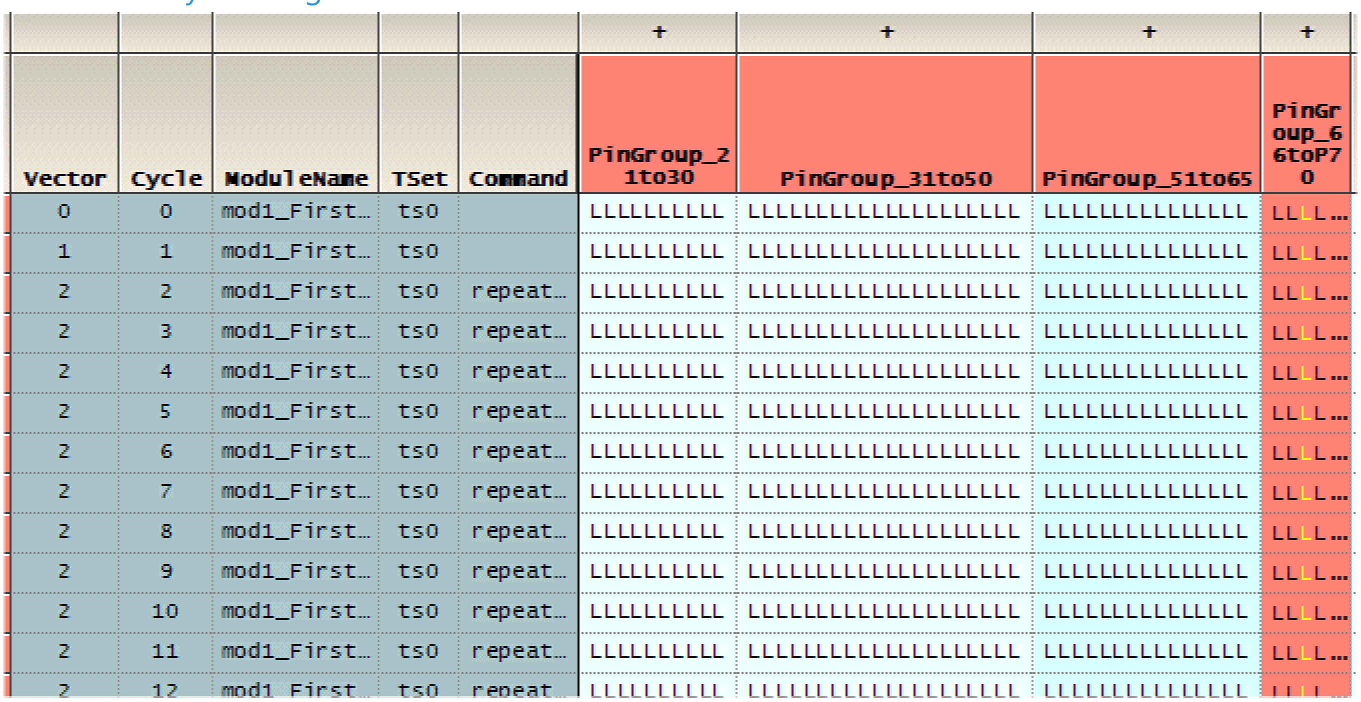
<	>

Displaying Digital HRAM : Viewing HRAM Data : Freezing and Unfreezing the View

---

Freezing and Unfreezing the View

When you are scrolling, you may want to keep some columns, such as the PatGen data, in view. You can do this by freezing the view. A solid dark line indicates where the view is frozen:



Vector	Cycle	ModuleName	TSet	Command	PinGroup_21to30	PinGroup_31to50	PinGroup_51to65	PinGroup_66toP70
					+	+	+	+
0	0	mod1_First…	ts0		LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
1	1	mod1_First…	ts0		LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	2	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	3	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	4	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	5	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	6	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	7	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	8	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	9	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	10	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	11	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…
2	12	mod1_First…	ts0	repeat…	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLL	LLLL…

Display with Columns Frozen

Select the column to the right of where you want to view frozen. You must select the entire column by clicking the column header. Then right-click and select Freeze View from the context menu. You can also use the Window>Freeze View command. The dark line is moved to the new location to the left of the selected column. All columns to the left of the line remain static during scrolling.

If you do not want the view frozen, use Window>Unfreeze View. The command is available on the right-click context menu as long as you do not have an entire column selected. No columns remain static during scrolling.

<	>

HRAMDisplayUsing.3.07

<	>

Displaying Digital HRAM : Viewing HRAM Data : Viewing Messages

Viewing Messages

Messages that do not require an immediate user response are written to the Message Panel. For details, see About the Message Panel.

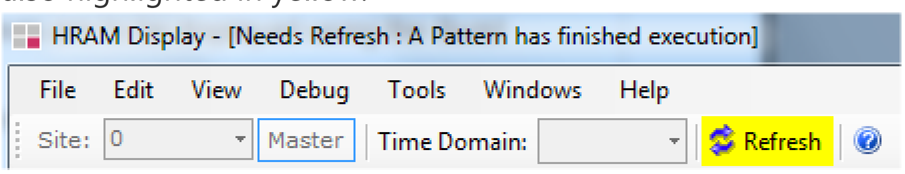
<	>

HRAMDisplayUsing.3.08

Displaying Digital HRAM : Viewing HRAM Data : Refreshing the Display

Refreshing the Display

If the current pattern has been reburst from outside HRAM Display, the message “Needs Refresh” is displayed in the window title bar, along with the reason the refresh is needed. The Refresh button is also highlighted in yellow:



Needs Refresh

Click this button or use View>Refresh or F12 to display the results of the reburst. If you reburst in HRAM Display, the display is automatically refreshed.

HRAMDisplayUsing.3.09

<	>

Displaying Digital HRAM : Viewing HRAM Data : Exporting HRAM Data

Exporting HRAM Data

The command File>Export HRAM Data exports the currently displayed HRAM data to a text file. The command brings up a browser that lets you specify the filename and folder location. The default folder is the one containing the workbook. The default filename is:

<patternname>_HramData_<timestamp>.txt

The ASCII file describes the symbols used to describe the HRAM data.

<	>

HRAMDisplayUsing.3.10

<	>

Displaying Digital HRAM : Viewing HRAM Data : Note on Keepalive (“?” Symbol)

Note on Keepalive (“?” Symbol)

If the cycles captured in HRAM occurred during a keepalive loop, then HRAM Display may show "?" in place of the patgen command because IG-XL automatically disables HRAM capture during keepalive. The VBT statement TheHdw.Digital.HRAM.PatGenInfo shows the same behavior.

<	>

HRAMDisplayUsing.3.11

<	>

Displaying Digital HRAM : Navigation

Navigation

The following navigation tools are available:

- Goto
- Viewstrips
- Scrollbars

<	>

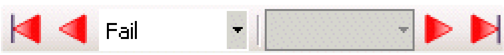
HRAMDisplayUsing.3.12

<	>

Displaying Digital HRAM : Navigation : Goto

Goto

The Goto control on the toolbar lets you go to a specific item in the display:



Goto Control

Use the drop-down list to select the type of information what you want to go: Fail, Cycle #, Text, or Pin. You can also use Edit>Select Search to select the type of information.

To specify which item of that type to go to (which cycle number or what text), the text box is enabled for you to enter or select a value. If you select Pin, the box becomes a drop-down listing the available pins.

The red buttons let you go (from left to right) to the first, the previous, the next, or the last instance of the specified item. If you are going to a specific item (for example, a specific cycle number) only the Next button is enabled; it takes you to the specified item.

The Edit menu has commands that go to the first, previous, next, or last instance. You can also use Ctrl plus the arrow buttons:

- First = Ctrl+Up
- Previous = Ctrl+Left
- Next = Ctrl+Right
- Last = Ctrl+Down

<	>

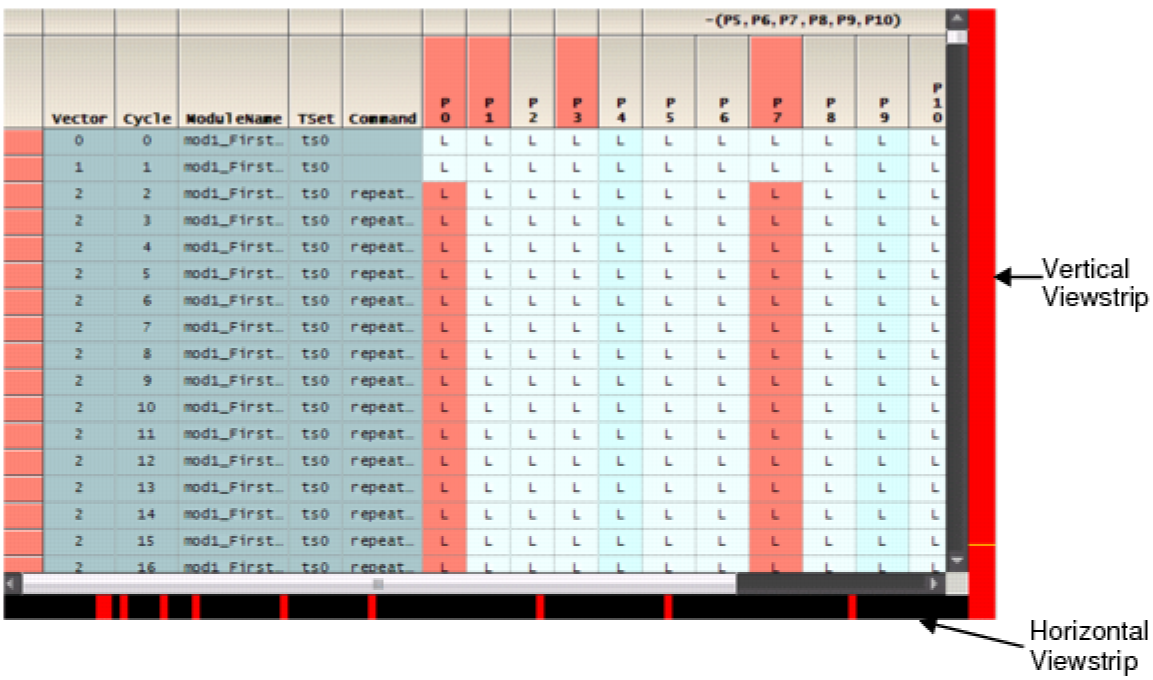
HRAMDisplayUsing.3.13

<	>

Displaying Digital HRAM : Navigation : Viewstrips

Viewstrips

If the data to be displayed extends beyond what is visible in the display area, a horizontal or vertical viewstrip or both become visible:



Display Showing Horizontal and Vertical Viewstrips

A viewstrip gives an overall view of the data, specifically of all captured failures across the horizontal time scale and the vertical vectors. It shows the relative positions of failures in relation to the entire span of data.

You can double-click at any point in a viewstrip to navigate to that position in the data. Clicking in the horizontal viewstrip scrolls only on the horizontal pin axis. Clicking in the vertical viewstrip scrolls only on the vertical vector axis.

The following colors are used to indicate items in the viewstrip:

- A red band indicates a failure at that location. If you hover over a failure on the horizontal viewstrip, a tooltip displays the fail pin name. If you hover over the vertical viewstrip, the vector number is displayed.

- The yellow line on the vertical strip indicates the current selected position.

<	>

HRAMDisplayUsing.3.14

<	>

Displaying Digital HRAM : Navigation : Scrollbars

Scrollbars

If the data to be displayed extends beyond the display area, a horizontal or vertical scrollbar or both appear at the right and bottom of the tool.

You can use the arrows at each end of the scrollbar to move smoothly in that direction. You can left-click the scrollbar to jump a small distance in that direction. If you right-click the scrollbar, the context menu gives you more options:

- Scroll here, that is, to the point at which you clicked on the scrollbar.
- Scroll all the way to the top or bottom or left edge or right edge.
- Move one page’s worth up or down or to the right or left.

When you are scrolling, you may want to keep certain columns, such as the PatGen data, in view. See Freezing and Unfreezing the View.

<	>

HRAMDisplayUsing.3.15

<	>

Displaying Digital HRAM : Burst Explorer and Properties

Burst Explorer and Properties

By default, the Burst Explorer and the Properties window are on the left side of the tool. If they are not visible, you can use commands on the View menu to make them visible.

The following topics are covered:

- Burst Explorer Treeview Structure
- Selecting Pins and PatGen Data
- Using View Setups
- Properties Window

<	>

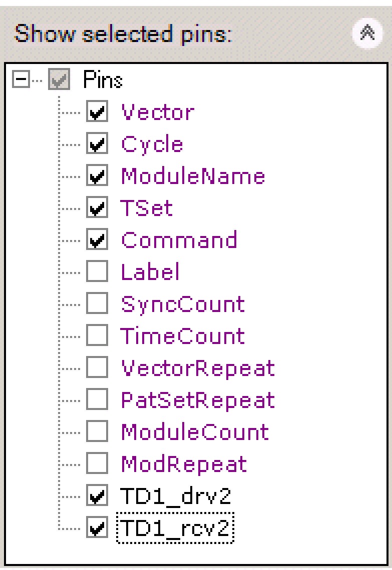
HRAMDisplayUsing.3.16

<	>

Displaying Digital HRAM : Burst Explorer and Properties : Burst Explorer Treeview Structure

Burst Explorer Treeview Structure

The Burst Explorer shows the pins, groups, and lists in the pattern, including items for PatGen data, shown in purple:



Burst Explorer

If there are multiple time domains, the time domains are the top node.

Each pin or pin group is a separate item under a time domain. Each pin group can be expanded.

Failing pins are in red, as are pin groups with failing pins in them.

Masked pins are shown in blue.

You can select which PatGen and pin data columns are displayed. See Selecting Pins and PatGen Data.

<	>

HRAMDisplayUsing.3.17

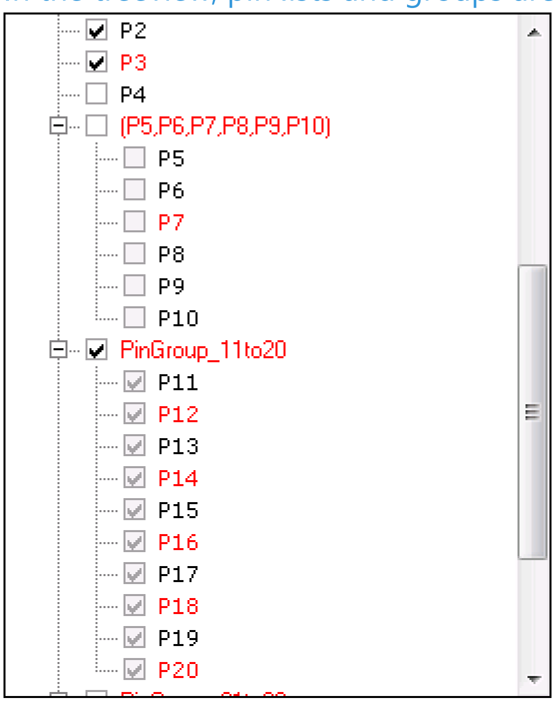
<	>

Displaying Digital HRAM : Burst Explorer and Properties : Selecting Pins and PatGen Data

Selecting Pins and PatGen Data

The checkboxes in the Burst Explorer control what PatGen data and pin data items are included in the display. To include an item in the display, check its box. To remove it, uncheck the box.

In the treeview, pin lists and groups are displayed as nodes with the individual pins below them:



Pin Lists and Groups

You can check or uncheck a pin list or group node, but not the individual pin nodes. This means that all the pins in a list or group are either selected or not selected. Checking the parent node checks all the children nodes with a light gray check mark.

By default, the Label PatGen column is not selected. If you select it, note that only global labels are displayed. Local labels are not displayed.

Clicking an item name itself does not select it, although it does show the item's properties in the Properties window. You must check the box to include it in the display.

Once you set up the display as you want it, you can save it to view setup. You can later recreate the display by selecting the setup. See Using View Setups.

<	>

HRAMDisplayUsing.3.18

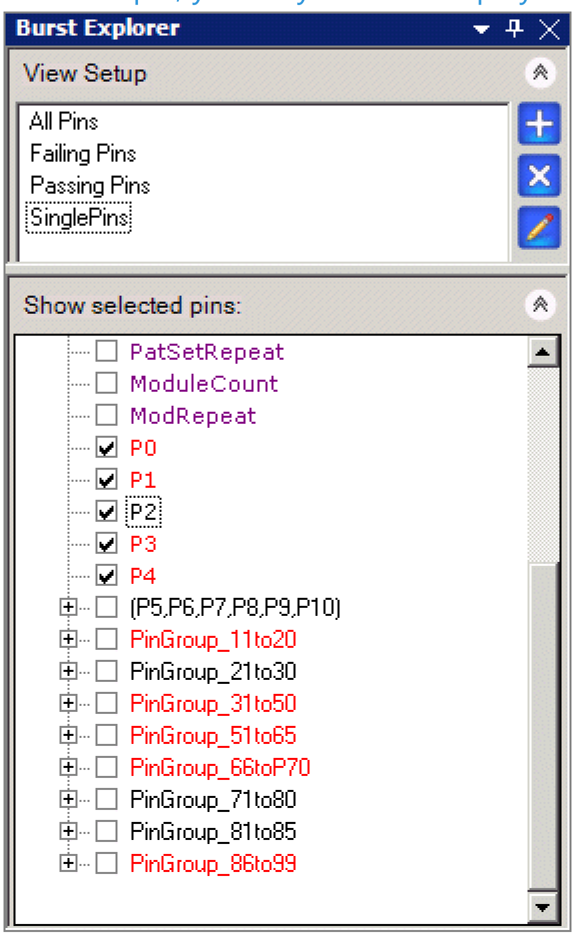
<	>

Displaying Digital HRAM : Burst Explorer and Properties : Using View Setups

Using View Setups

You can use the Burst Explorer to create view setups. A view setup lets you select the items that you want included in the rows of the display and save the selection as a named setup. You can later redisplay those rows by clicking the name of the setup.

For example, you may want to display only single pins:



Creating View Setups in Burst Explorer

The available setups are listed in the View Setup panel. If this is not visible, use View>Burst Explorer to make it visible.

The default built-in setups are All Pins, Failing Pins, and Passing Pins. You cannot modify or delete this setup.

 To create a view setup, select the items that you want included by checking their boxes in the Burst Explorer. Then click the Add button. A new string is added to the View Setup panel with the default name “setupN.” The name is in edit mode, so you can change it immediately.

Once you add a setup, you can double-click the name to display the selected rows.

 To delete a setup, select it in the View Setup panel and click the Delete button. To delete all user-created setups, right-click in the white area of the View Setup panel (not on a setup name) and select Delete All.

 To modify a setup, select it in the View Setup panel. Make changes to the check boxes. Then click the Update button.

To rename a setup, select it in the View Setup panel, right-click, and select Rename. Note that Apply, Delete, and Update are also on the right-click menu.

<	>

HRAMDisplayUsing.3.19

<	>

Displaying Digital HRAM : Burst Explorer and Properties : Properties Window

Properties Window

The Properties window displays the properties of the currently selected pin.

If the Properties window is not visible, use View>Properties to display it.

The Properties window can be divided into categories. The first category is always General. The other categories depend on the kind of item selected.

You can specify how the properties are to be displayed using the buttons at the top of the Properties window:

- Categorized - for example, General, Levels, Timing
- Alphabetical, without category dividers

The Properties window displays the properties of the selected pin, pin group, or pin list.

You can modify some properties before rebursting to perform basic debug. Properties that you can change are in black. Properties that you cannot change are grayed out. For example, you can change a pin's voltage levels, but not its name. You can change a pin's timing edge or period, but not its timeset or edgeset name.

Any pin can be masked. The Mask property is set to True or False. You can mask or unmask pin by changing the value in the Properties window. You can also select the pin or group in the Burst Explorer and select Debug>Mask. See Masking Pins.

Which pin properties are displayed depends where you select a pin: in the Burst Explorer or in the display area. If you select a pin in the Burst Explorer, the Properties window displays levels information. If you select a pin in the pin headers column, it is the same as selecting it in the Burst Explorer. If you select a point of a pin lane in the waveform display area, you are actually selecting one or more cycles. The associated timeset is available, so the Properties window can display timing and format information and pattern execution information, in addition to levels information. Some timing and format values can be changed for debugging. Selecting a pin, group, or list in the row headers is the same as selecting it in the Burst Explorer: you have not selected cycles, so only levels information is displayed. You can select cycles only for pins, not for pin groups or lists.

If you select a range that includes multiple pins or cycles, the Properties window displays information for the rightmost cycle and bottommost pin. If you select discontinuous cycles, the last cycle selected is displayed.

The following paragraphs describe the kinds of properties displayed for each selection, but does not list all properties.

If you select a pin, pin group, or pin list in the Burst Explorer, the following is displayed in the Properties window:

General	Pin name, mask state (which you can change), instrument
Levels	The pin’s voltage levels. You can modify the levels.

If you select a pin in the waveform display, the following is displayed in the Properties window:

General	Pin name, cycle and vector number, fail state, module name, but not the mask state. You cannot change any of this information.
Levels	The pin’s voltage levels. You can modify the levels.
Timing/Format	All time set and edge set information, plus init state and start state. You can modify the timing edges, periods, and init and start states, but not items such as data format or pin setup.

<	>

HRAMDisplayUsing.3.20

<	>

Displaying Digital HRAM : Bursting Patterns

Bursting Patterns

You can reburst the most recently burst pattern or pattern set. This lets you debug the patterns.

The following topics are covered:

- Burst Commands
- Masking Pins
- Virtual HRAM Size

<	>

HRAMDisplayUsing.3.21

<	>

Displaying Digital HRAM : Bursting Patterns : Burst Commands

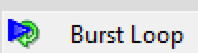
Burst Commands

Burst commands are on the Debug menu. In addition, they are on the toolbar. By default, the Burst command appears on the button:

Default Burst Button

If you click the down arrow on the right of the button, another command appears. Select any of the commands. The last selected command then appears on the button.

The Burst commands are as follows:

	Rebursts the last burst patterns or pattern sets.
	Rebursts in a loop. Use Halt to stop the loop. The status bar Burst Count indicates the number of times the pattern has been burst.
	Rebursts in a loop, until the first pass on an HSD pin is detected. The status bar Burst Count indicates the number of times the pattern has been burst. You can use Halt to stop the loop.
	Rebursts in a loop until the first failure on an HSD pin is detected. If no failures are detected, you can use the Halt button or command to stop rebursting. The Burst Count, the number of times the pattern was burst, is displayed in the status bar.

Bursting rebursts the last burst patterns or pattern sets in the job. This could be different from the ones currently viewable in the tool.

HRAM Display automatically refreshes after the reburst.

While a pattern is bursting, the Halt button is enabled, letting you halt the burst.

The test instance that bursts the pattern as part of the job may specify an interpose function to be executed before the pattern burst (PrePatF) or after the pattern burst (PostPatF). When you burst these patterns in PatternTool, it may be desirable to execute one or both of these interpose functions as well. PatternTool lets you enable or disable the execution of one or both functions:

	Debug>PrePatF Enable or disable the execution of the PrePatF function before each pattern burst.
	Debug>PostPatF Enable or disable the execution of the PostPatF function after each pattern burst.

Note that these settings are uniform across PatternTool, Waveform Display, and HRAM Display. Changing a setting in one tool changes it for the others as well. In this way, pattern reburst is the same across all tools.

<	>

HRAMDisplayUsing.3.22

<	>

Displaying Digital HRAM : Bursting Patterns : Masking Pins

Masking Pins

You can mask a pin before reburstng. Masking failures is a common debug activity that lets you concentrate on a subset of failures of interest.

To mask a pin, select it in the Burst Explorer. Then right-click and select Mask from the context menu. The command Debug>Mask is also enabled when you select a pin in the Burst Explorer.

You can select and mask an entire pin group or list or a single pin in the group or list.

When a pin is masked, it is displayed in blue in the Burst Explorer. If you mask a pin in a pin group or list, the group or list name is also displayed in blue.

<	>

HRAMDisplayUsing.3.23

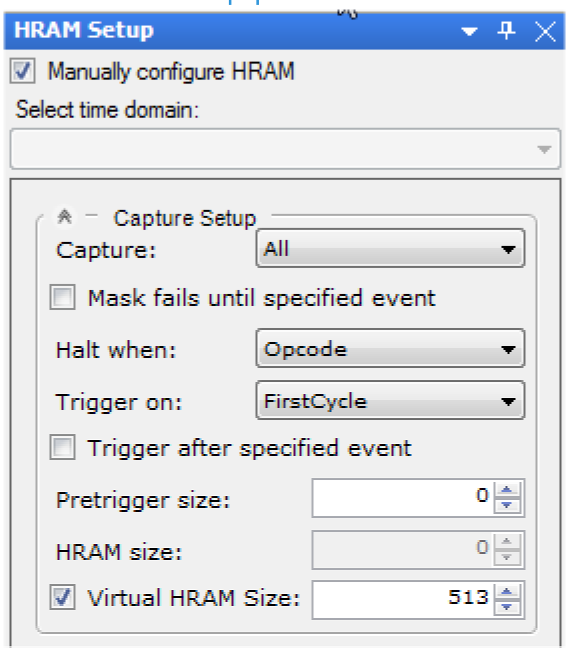
<	>

Displaying Digital HRAM : Bursting Patterns : Virtual HRAM Size

Virtual HRAM Size

Each digital instrument has a maximum number of cycles that it can capture in HRAM. HRAM Display supports a virtual HRAM that lets you capture more than this maximum number of cycles.

The HRAM Setup pane has this checkbox:



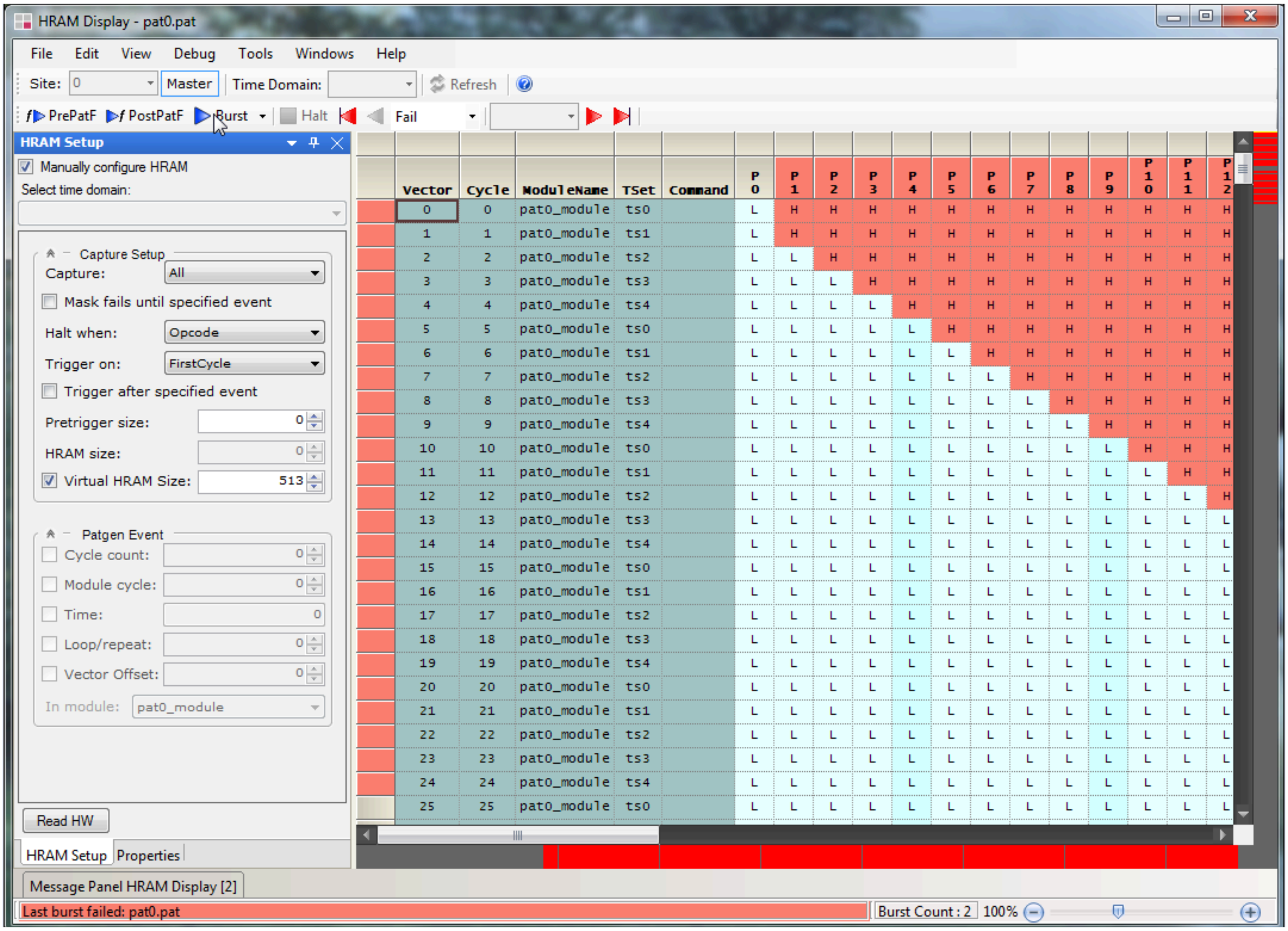
Virtual HRAM Size Checkbox

Check the box. Note that this disables the HRAM size control.

Enter the number of cycles you want to capture. If timing is set to By16 mode, the number must be a multiple of 16. Otherwise, it can be any number.

Virtual HRAM is supported only for the digital instruments (HPM, HSD1000, UltraPin800, UltraPin1600, and UltraPin4000).

When you reburst the pattern, HRAM Display bursts the pattern as many times as it takes to accumulate the specified number of cycles:



Virtual HRAM Display

For example, this pattern contains 512 non-repeating vectors. Because the virtual HRAM size is set at 513, it must burst the pattern a second time to get the 513 cycles. The burst count in the status bar is therefore set to 2.

On the first burst, HRAM Display uses the current HRAM settings: for example, starting after a certain event or capturing a number of pretrigger cycles. In rebursting the pattern, HRAM Display temporarily overrides the HRAM setup so it can start the second burst at the next cycle after the end of the previous burst.

The virtual HRAM is captured only in HRAM Display, not in the HRAM hardware. The other digital tools, such as PatternTool and Waveform Display, can see only the actual captured HRAM, not the virtual HRAM. If you reburst the pattern from one of these other tools, HRAM Display loses the virtual HRAM display.

Once you have burst several times using virtual HRAM, you can export the data displayed to a text file using File>Export HRAM Data. Note that this exports all accumulated data, not only the last burst.

<	>